25ECPCC305IAE-I

Seat No. : EU1257055

7 AUG 2026

Student Name : Harsh

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

Internal Assessment Examinations, August 2026

B.Tech. Electronics & Computer Science - SEMESTER III

25ECPCC305 - Data Structure & Algorithms using Python

Time : 1 Hour **Maximum Marks : 20**

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Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max marks(5)	Course Outcome	BT Level
	a)Which data structure is immutable in Python?	1	CO1	L1
	b)Which Python module provides the deque data structure?	1	CO1	L1
	c)Which technique processes consecutive elements efficiently?	1	CO2	L1
	d)Which strategy solves a problem by dividing it into smaller subproblems?	1	CO2	L1
	e)Which operation removes an element from the right end of a deque?	1	CO3	L2
	f)Why are queues used for scheduling?	1	CO3	L2
Q2	a)Calculate time complexity and space complexity for following Python code: n = int(input("Enter the value of n: ")) for i in range(n): for j in range(i + 1): print("*", end=" ") print()	5	CO1	L2
	OR			
	b)Explain the concept of recursion with a suitable example.	5	CO1	L2
Q3	a)Implement the N-Queens problem using Backtracking.	5	CO2	L3
	OR			
	b)Compare the complexity of Linear Search and Binary Search using suitable examples.	5	CO2	L3
Q4	a)Write and analyse a Python program to insert nodes at different positions in a singly linked list	5	CO3	L4
	OR			

b)Develop and analyse a Python program to rotate an array by k positions.	5	CO3	L4

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